

DEFINITION — HOW DATA IS DESCRIBED

CONTRACTS6



ODCS  
BITOL @ LF



OpenAPI  
LF



AsyncAPI  
LF



GraphQL  
LF



gRPC  
CNCF



OData  
OASIS

DATA PRODUCTS4



ODPS  
BITOL @ LF



DPDS  
ODM



ODPSpec  
LF



DPROD  
OMG

SCHEMA5



XML Schema  
W3C



JSON Schema  
IETF



SQL DDL  
ISO/IEC



AVRO Schema  
ASF



Protobuf  
GOOGLE

SEMANTICS6



OSI  
OSI INITIATIVE



RDF/OWL  
W3C



DCAT  
W3C



SKOS  
W3C



SHACL  
W3C



JSON-LD  
W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS7



CSV  
IETF



JSON  
IETF



XML  
W3C



PARQUET  
ASF



AVRO  
ASF



ORC  
ASF



Lance  
LANCE

OPEN TABLE FORMATS4



Iceberg  
ASF



Delta  
LF



Hudi  
ASF



Lance  
LANCE

STORAGE SYSTEMS2



S3  
AWS



HDFS  
ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY3



JDBC  
JCP



ODBC  
ISO/IEC



ADBC  
ASF

INTERCONNECTION3



HTTP  
IETF



FTP / SFTP  
IETF



Delta Sharing  
LF

MESSAGING4



Kafka  
ASF



MQTT  
OASIS



CloudEvents  
CNCF



AMQP  
OASIS

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT2



DataFrame API  
DATA-APIS



Apache Arrow  
ASF

PROCESSING6



dbt  
DBT LABS



Spark  
ASF



Beam  
ASF



Pandas  
NUMFOCUS



Ibis  
IBIS



SQL DML  
ISO/IEC

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS4



Iceberg Catalog  
ASF



Unity Catalog  
LF



Hive Metastore  
ASF



Schema Registry  
CONFLUENT

LINEAGE1



OpenLineage  
LF

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY2



SQL  
ISO/IEC



SPARQL  
W3C

DATA QUALITY3



dbt  
DBT LABS



Great Expectations  
GX LABS



SodaCL  
SODA

OBSERVABILITY2



Open Telemetry  
CNCF



OORS  
BITOL @ LF

POLICIES1



OPA  
CNCF

AI INTERFACES2



MCP  
LF



A2A  
LF